

Grade 7
Unit 4
Lesson 9 Practice

Name _____
Date _____
Period _____

For questions 1 – 10, round to the nearest hundredth. Use 3.14 for π .

Given the radius, determine the area of each circle.

1. 8in

$$\begin{aligned}A &= \pi r^2 \\A &= 3.14 \cdot 8^2 \\A &= 3.14 \cdot 64 \\A &= 200.96 \text{ in}^2\end{aligned}$$

2. 32.4cm

$$\begin{aligned}A &= \pi r^2 \\A &= 3.14 \cdot (32.4)^2 \\A &= 3.14 \cdot 1,049.79 \\A &= 3,296.2464 \\A &= 3,296.25 \text{ cm}^2\end{aligned}$$

3. $\frac{4}{5}\text{ft}$

$$\begin{aligned}A &= \pi r^2 \\A &= 3.14 \cdot \left(\frac{4}{5}\right)^2 \\A &= 3.14 \cdot \frac{16}{25} \\A &= 3.14 \cdot 0.64 \\A &= 2.0096 \\A &= 2.01 \text{ ft}^2\end{aligned}$$

Given the diameter, determine the area of each circle.

4. 18m

$$\begin{aligned}r &= d \div 2 \\r &= 18 \div 2 \\r &= 9 \\A &= \pi r^2 \\A &= 3.14(9)^2 \\A &= 3.14 \cdot 81 \\A &= 254.34 \text{ m}^2\end{aligned}$$

5. 76.1mm

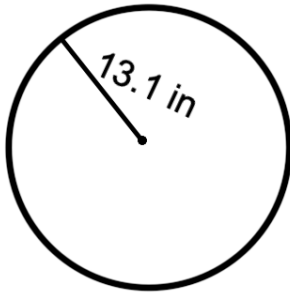
$$\begin{aligned}r &= d \div 2 \\r &= 76.1 \div 2 \\r &= 38.05 \\A &= \pi r^2 \\A &= 3.14(38.05)^2 \\A &= 3.14 \cdot 1,447.8025 \\A &= 4,546.10 \text{ mm}^2\end{aligned}$$

6. 19.1in

$$\begin{aligned}r &= d \div 2 \\r &= 19.1 \div 2 \\r &= 9.55 \\A &= \pi r^2 \\A &= 3.14 \cdot (9.55)^2 \\A &= 3.14 \cdot 91.2025 \\A &= 286.38 \text{ in}^2\end{aligned}$$

Given the radius or diameter, find the area of the circle.

7.



$$A = \pi r^2$$

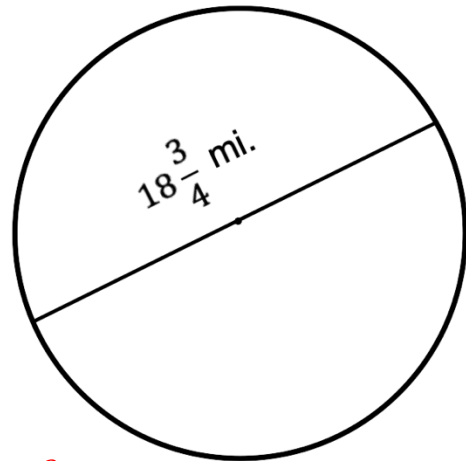
$$A = 3.14(13.1)^2$$

$$A = 3.14(171.61)$$

$$A = 538.8554$$

$$A = 538.86 \text{ in}^2$$

8.



$$d = 18\frac{3}{4}$$

$$r = d \div 2$$

$$r = 18\frac{3}{4} \div 2$$

$$r = 9.375$$

$$A = \pi r^2$$

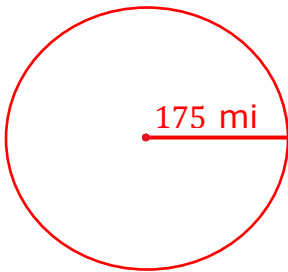
$$A = 3.14(9.375)^2$$

$$A = 3.14(87.890625)$$

$$A = 275.98 \text{ mi}^2$$

When an earthquake occurs, shaking can be felt in every direction from the epicenter. An earthquake in California was felt by everyone within 175 miles from the epicenter.

9. Draw a diagram of the scenario.



10. Determine the area that felt the earthquake.

$$A = \pi r^2$$

$$A = 3.14(175)^2$$

$$A = 3.14(30,625)$$

$$A = 96,162.50 \text{ mi}^2$$